



1.3

Rates of Change in Linear and Quadratic Functions

Core-to-Challenge Worksheet and AP Practice

AP Precalculus - Unit 1 | ECHS Mathematics | Teacher: Mohammad Abu-Ghuwaleh

Student Information

Name: _____ Class: _____ Date: _____

Directions

Show mathematical evidence. Retain domain restrictions, label graphs, include units, and write complete interpretations. Calculator use is permitted only when the item explicitly requires approximation, regression, or numerical solving.

Readiness

1. Write one precise sentence explaining this principle: Equal input steps are essential when using finite differences.

2. Write one precise sentence explaining this principle: Linear functions have constant first differences and zero second differences.

3. Write one precise sentence explaining this principle: Quadratic functions have constant nonzero second differences.

4. Write one precise sentence explaining this principle: For $f(x) = ax^2 + bx + c$ with step size h , the constant second difference is $2ah^2$.

Core

5. The values 4, 9, 14, 19, 24 occur at consecutive integers. Classify the pattern.
6. The values 2, 7, 16, 29, 46 occur at consecutive integers. Determine the quadratic leading coefficient.
7. Explain why a quadratic can have the same average rate on two different intervals.
8. State what the formula $\Delta f = f(x + h) - f(x)$ tells you and name any required restriction.
9. State what the formula $\Delta^2 f = \Delta f(x + h) - \Delta f(x)$ tells you and name any required restriction.
10. State what the formula $\Delta^2(ax^2 + bx + c) = 2ah^2$ tells you and name any required restriction.

Medium

11. Classify the data at consecutive integer inputs: 2, 7, 12, 17, 22. Show differences.
12. Classify the data at consecutive integer inputs: 1, 4, 9, 16, 25. Show differences.
13. Classify the data at consecutive integer inputs: 5, 1, -3, -7, -11. Show differences.
14. Classify the data at consecutive integer inputs: 3, 8, 17, 30, 47. Show differences.

15. Classify the data at consecutive integer inputs: 0, 2, 8, 18, 32. Show differences.

Hard

16. A quadratic table at step size 2 has constant second difference 24. Find its leading coefficient.

17. For $f(x) = 2x^2 - 5x + 1$, find AROC on $[x, x + 1]$ and describe its pattern.

18. An arithmetic sequence and a quadratic sequence agree at $n = 0, 1$. Must they agree for all n ?

19. Construct a quadratic sequence with first values 4, 1, 2.

Difficult

20. Prove that for $f(x) = ax^2 + bx + c$, AROC on $[p - h, p + h]$ equals $2ap + b$.

21. A table at inputs 0, 2, 4, 6 has outputs 7, 15, 31, 55. Find a quadratic formula.

22. Can unequal input spacing produce constant raw first differences for a nonlinear function? Give an example.

Challenge / HOT

23. Show that a sequence with constant nonzero second differences cannot be arithmetic.

24. Determine all quadratics whose AROC on every interval centered at 3 is 10.

AP-Style Multiple Choice

25. Constant nonzero second differences indicate
 (A) linear
 (B) quadratic

- (C) exponential
(D) rational
26. For unit steps, the second difference of $3x^2 + 2x$ is
(A) 3
(B) 5
(C) 6
(D) 9
27. A linear function has
(A) constant first differences
(B) constant ratios
(C) constant nonzero second differences
(D) variable slope
28. Which list is quadratic?
(A) 2, 5, 8, 11
(B) 1, 2, 4, 8
(C) 2, 6, 12, 20
(D) 5, 5, 5, 5
29. If step size doubles, the quadratic second difference is multiplied by
(A) 2
(B) 4
(C) 8
(D) unchanged
30. For x^2 , AROC on $[a, a + 1]$ is
(A) a
(B) $2a$
(C) $2a + 1$
(D) $a^2 + 1$

AP-Style Free Response

31. A table at x-values $-2, -1, 0, 1, 2$ has y-values 19, 8, 3, 4, 11. (a) Use differences to identify the family. (b) Find a formula. (c) Determine the interval where the unit-step average rates change sign.

32. Let $Q(x) = ax^2 + bx + c$. The AROC on $[0, 2]$ is 5 and on $[2, 4]$ is 13, with $Q(0) = 1$. Find Q and explain the second-difference connection.

Reflection

Identify one item where a domain restriction, unit, assumption, or representation changed your reasoning. Explain in 2-3 sentences.

Original ECHS questions. Selected concepts are informed by Pearson and Holt Precalculus; no textbook exercises are reproduced.